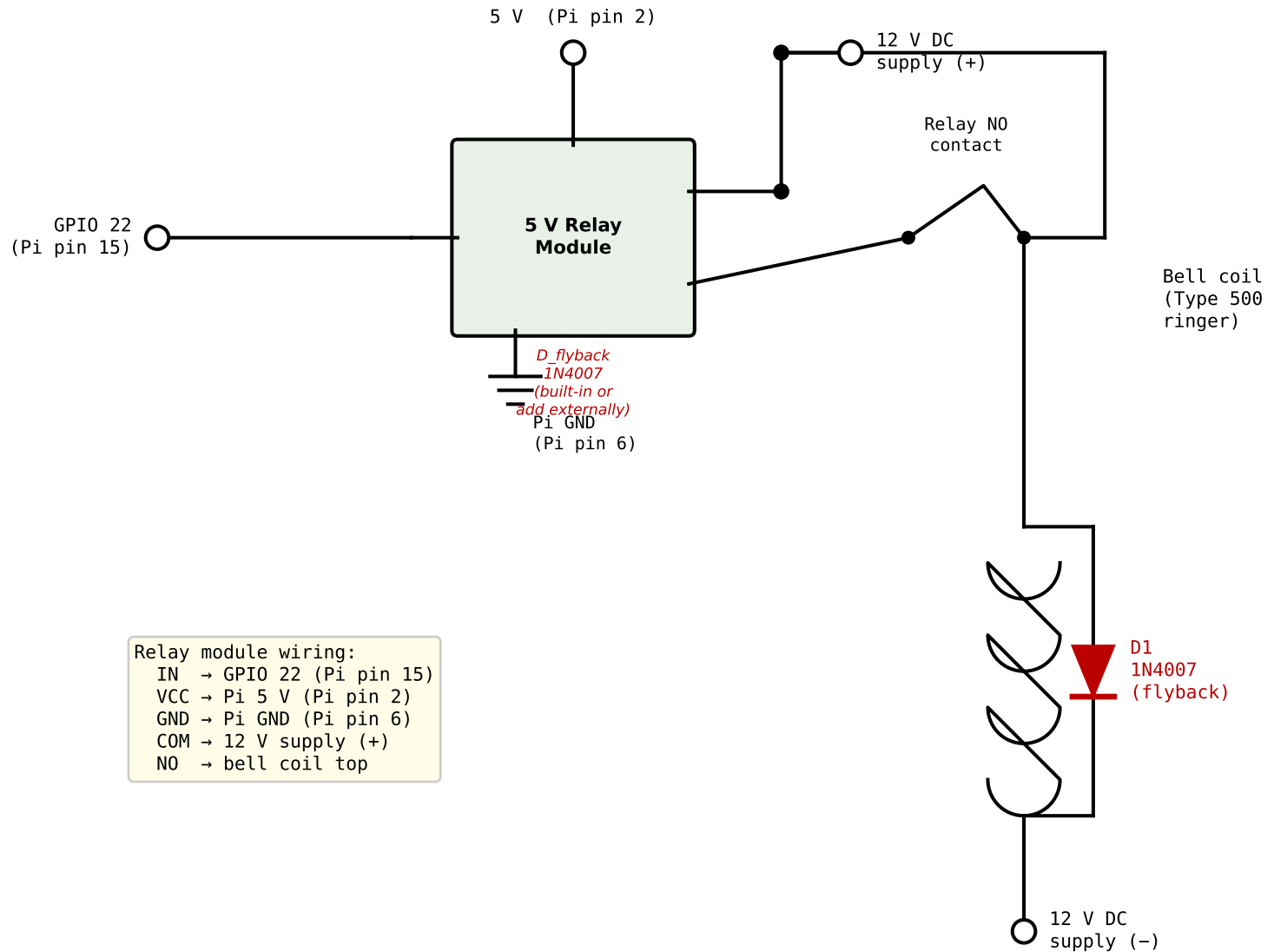


Bell Wiring — Option 1: Simple Relay

GPIO 22 → 5 V Relay Module → Bell Coil (single strike per toggle)



Relay module wiring:
IN → GPIO 22 (Pi pin 15)
VCC → Pi 5 V (Pi pin 2)
GND → Pi GND (Pi pin 6)
COM → 12 V supply (+)
NO → bell coil top

Operation: GPIO 22 HIGH → relay energises → 12 V reaches bell coil → clapper strikes ONCE per toggle.
The script's 2 s ON / 4 s OFF cadence produces two strikes per ring cycle (one energise, one de-energise).
For authentic mechanical oscillating ring use Option 2 (555 timer at 20 Hz) instead.
Relay module: 5 V active-high board (SRD-05VDC-SL-C, HW-482, or equivalent).
D1 (red) – flyback diode: ESSENTIAL – clamps inductive back-EMF that would destroy the transistor inside the relay module.
Bell supply: 12–24 V DC, ≥ 500 mA. Higher voltage rings louder; do not exceed bell coil rating.
Pi GPIO 22 max output current: 16 mA – NEVER connect a relay coil directly to a GPIO pin; always use a relay module.